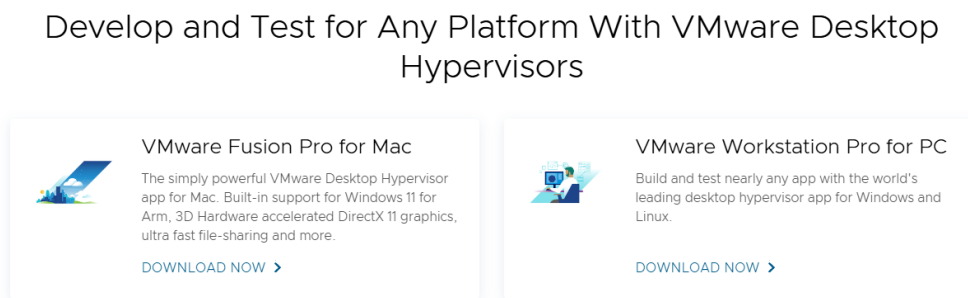


操作系统实验环境配置教程

一、下载虚拟机 VMware Workstation pro 17

方法一：通过官方网站下载（有些注册的问题比较麻烦）

<https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>



方法二：从超星泛雅上获取

安装过程需要注意的事项：修改默认安装路径，建议不要放到 C 盘

二、下载 Ubuntu 22.04

方法一：通过官方网站下载

<https://releases.ubuntu.com/jammy/>



Select an image

Ubuntu is distributed on three types of images described below.

Desktop image

The desktop image allows you to try Ubuntu without changing your computer at all, and at your option to install it permanently later. This type of image is what most people will want to use. You will need at least 1024MiB of RAM to install from this image.

64-bit PC (AMD64) desktop image

Choose this if you have a computer based on the AMD64 or EM64T architecture (e.g., Athlon64, Opteron, EM64T Xeon, Core 2). Choose this if you are at all unsure.

方法二：从超星泛雅上获取

镜像文件大约 4.5GB，下载会比较慢，可以通过 U 盘从下好的同学那里拷贝

三、在 VMware 上部署 Ubuntu

1. 打开 VMware，选择创建新的虚拟机，选择“典型”，下一步



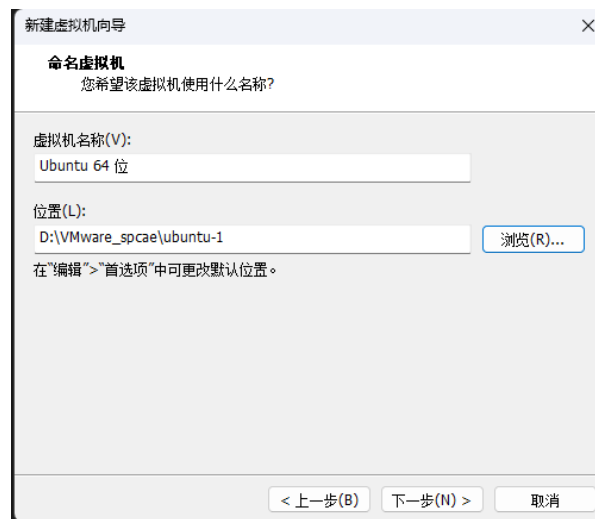
2. 安装下载路径填写 Ubuntu 镜像的路径，下一步



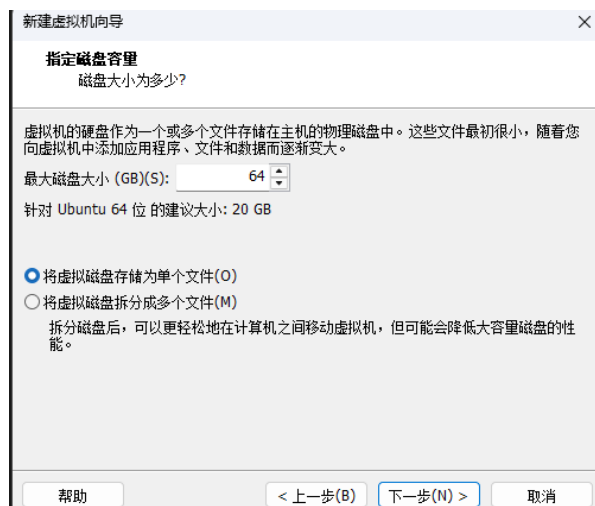
3. 填写信息，注意使用英文，密码建议设置的简单一些（如 root），密码一定不能忘记



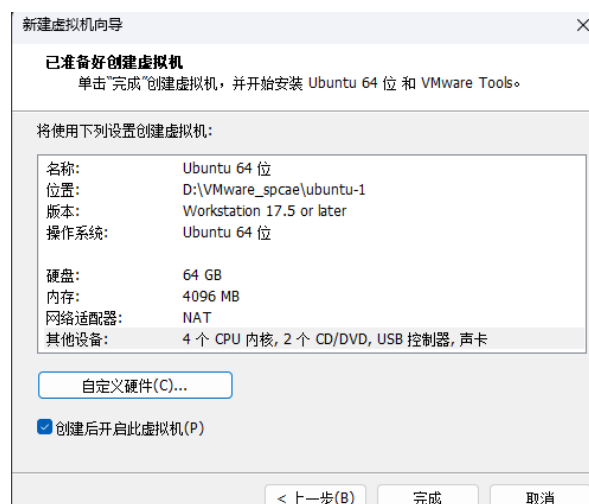
4. 选择操作系统的安装路径（注意不要用默认的 C 盘路径）



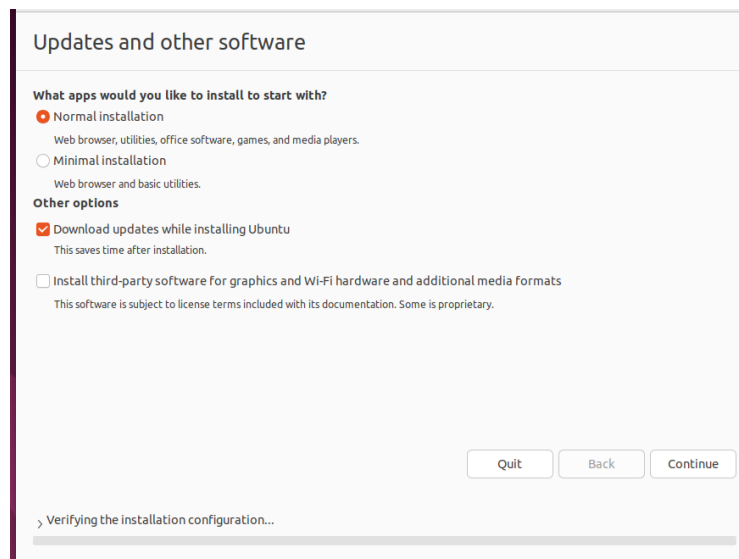
5. 配置磁盘大小（64GB 足够）



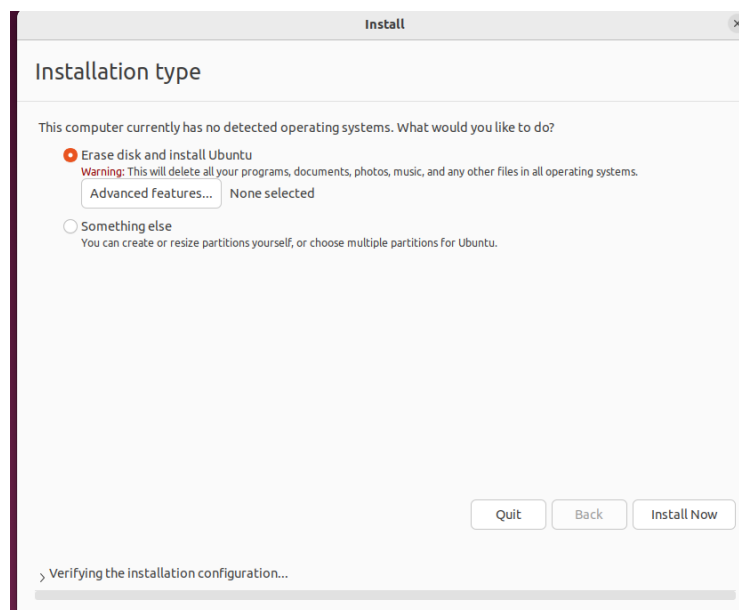
6. 选择自定义硬件，按照电脑配置高低选择内存和 CPU 配置。默认 4GB+2 个 CPU 内核，建议改成 4GB + 4(2*2)CPU 内核 或 8GB + 8(2*4)CPU 内核。点击完成。



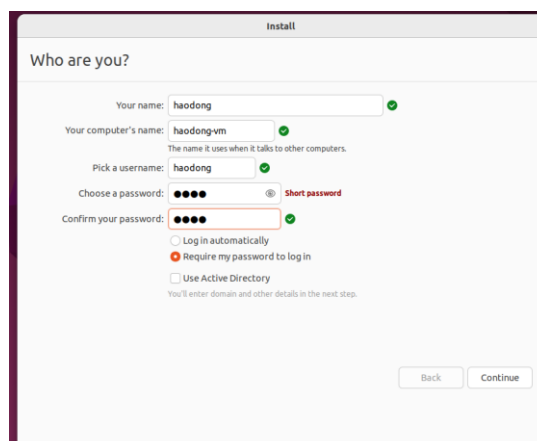
7. 虚拟机启动后选择中文，然后进入下面的页面，保持默认选项。



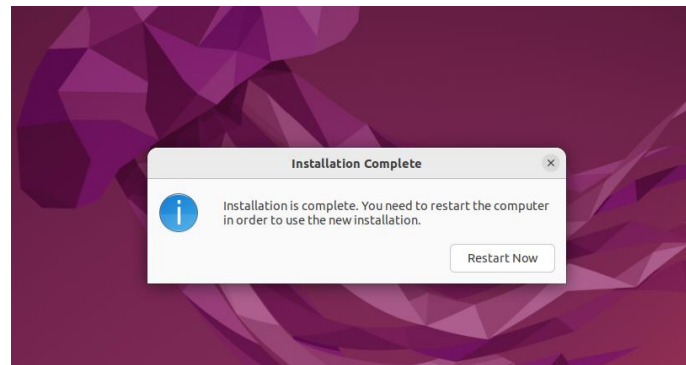
8. 保持默认选项，点击 Install Now



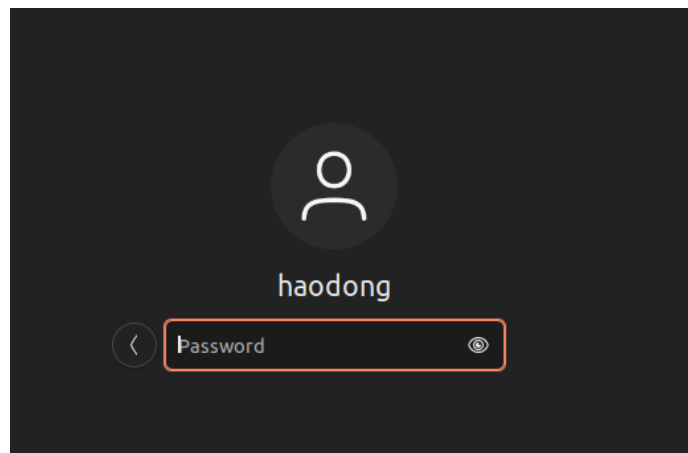
9. 选择时区，填写登陆信息（建议和之前保持一致），点击 continue，等待安装



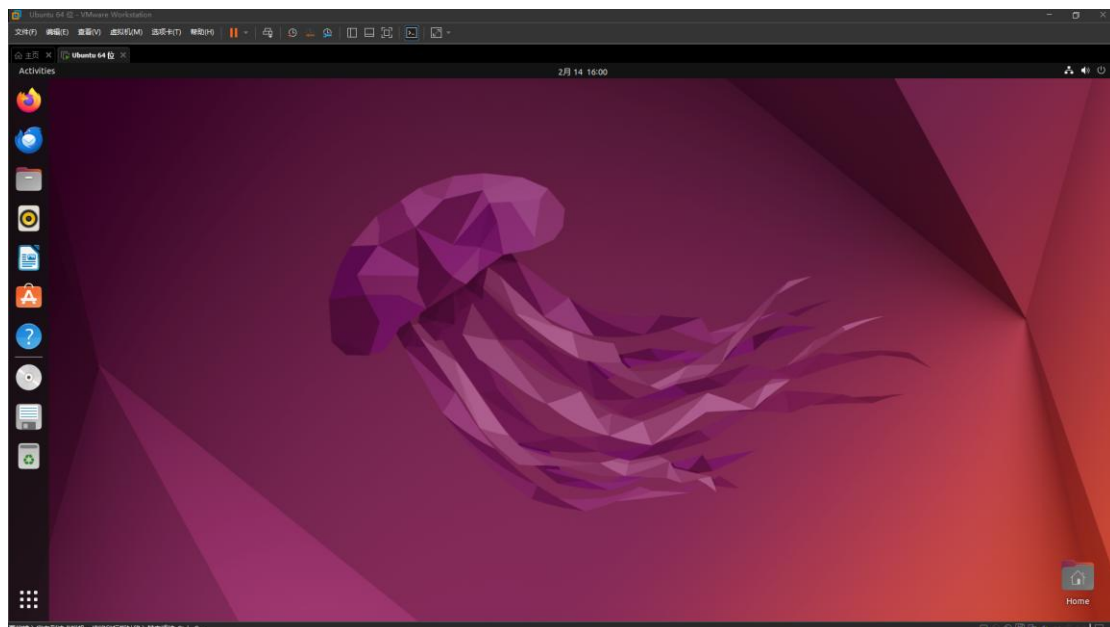
10. 安装完成后重启虚拟机



11. 点击用户名，输入密码登陆（前面设置的密码）



12. 进入系统，弹出的窗口都跳过或默认即可，拒绝系统升级（最好不要升级到 Ubuntu 24.04）



四、安装工具链和克隆代码

1. 在 Ubuntu 环境安装 vscode 和 git

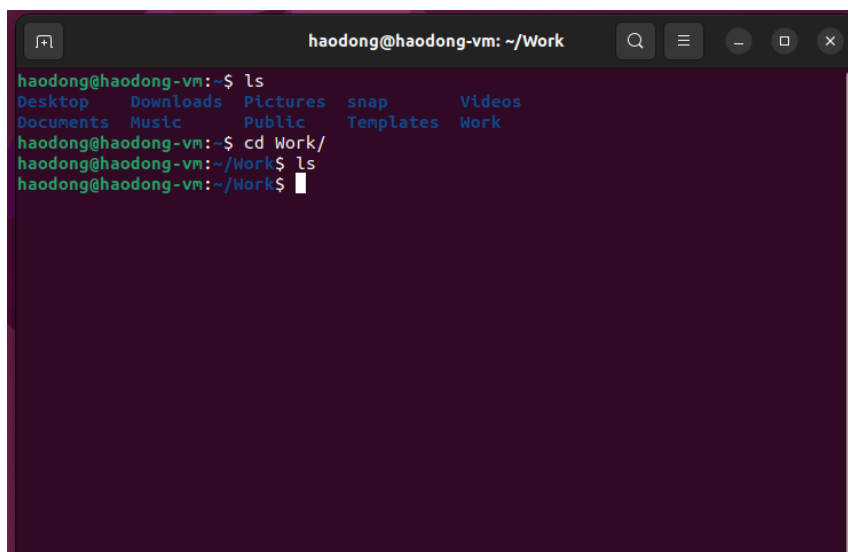
Ctrl + Alt + T 快捷打开终端窗口

- (1) 输入 `mkdir Work` 创建 Work 目录作为实验目录，输入 `ls` 查看是否创建成功
- (2) 输入 `sudo snap install code --classic` 安装 vscode。sudo 代表高级权限，可能要求输入密码后执行，输入密码的过程中密码不会显示在屏幕上，输入完成点击回车。
- (3) 输入 `code Work` 通过 vscode 打开 Work 目录，确认 vscode 能正常工作，后面的实验建议使用 vscode 完成。(vscode 的环境可以通过账号登陆和同步与 Windows 保持一致，也可以自己安装插件配置环境，vscode 开发环境配置的部分请自行上网查阅资料)
- (4) 在命令行里输入 `sudo apt install git` 安装 Linux 下使用的 git，遇到 [Y/n] 的问题都输入 Y，需要输密码的地方输入密码。
- (5) 可以通过 `git --version` 和 `code -v` 查看是否安装成功(版本不一定非要是下图所示)

```
haodong@haodong-vm:~$ git --version
git version 2.34.1
haodong@haodong-vm:~$ code -v
1.97.2
e54c774e0add60467559eb0d1e229c6452cf8447
x64
haodong@haodong-vm:~$
```

2. 下载环境依赖和 QEMU

- (1) `cd Work` 进入 Work 目录，输入 `ls` 确认 Work 目录下此时是空的



A terminal window titled 'haodong@haodong-vm: ~/Work' showing the following commands and output:

```
haodong@haodong-vm:~$ ls
Desktop  Downloads  Pictures  snap      Videos
Documents Music      Public   Templates Work
haodong@haodong-vm:~$ cd Work/
haodong@haodong-vm:~/Work$ ls
haodong@haodong-vm:~/Work$
```

(2) 输入 `sudo apt-get install build-essential gdb-multiarch qemu-system-misc gcc-riscv64-linux-gnu binutils-riscv64-linux-gnu` 安装过程中遇到的 [Y/n] 输入 Y

(3) 依次输入下面的命令

```
wget https://download.qemu.org/qemu-5.1.0.tar.xz
```

```
tar xf qemu-5.1.0.tar.xz
```

```
rm -rf qemu-5.1.0.tar.xz
```

```
ls
```

此时 Work 目录下应该只有 qemu-5.1.0 的目录

```
haodong@haodong-vm:~/Work$ rm -rf qemu-5.1.0.tar.xz
haodong@haodong-vm:~/Work$ ls
qemu-5.1.0
haodong@haodong-vm:~/Work$
```

(4) 输入 `./configure --disable-kvm --disable-werror --prefix=/usr/local --target-list="riscv64-softmmu"` 并不断尝试和解决报错 (错误可以丢给 AI 问原因, 一般可以通过安装一些包来解决)

```
haodong@haodong-vm:~/Work/qemu-5.1.0$ ./configure --disable-kvm --disable-werror --p
refix=/usr/local --target-list="riscv64-softmmu"

ERROR: pkg-config binary 'pkg-config' not found

haodong@haodong-vm:~/Work/qemu-5.1.0$ sudo apt-get install pkg-config
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  pkg-config
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 48.2 kB of archives.
After this operation, 134 kB of additional disk space will be used.
Get:1 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy/main amd64 pkg-config amd64 0
.29.2-1ubuntu3 [48.2 kB]
Fetched 48.2 kB in 1s (60.2 kB/s)
Selecting previously unselected package pkg-config.
(Reading database ... 211235 files and directories currently installed.)
Preparing to unpack .../pkg-config_0.29.2-1ubuntu3_amd64.deb ...
Unpacking pkg-config (0.29.2-1ubuntu3) ...
Setting up pkg-config (0.29.2-1ubuntu3) ...

haodong@haodong-vm:~/Work/qemu-5.1.0$ ./configure --disable-kvm --disable-werror --p
refix=/usr/local --target-list="riscv64-softmmu"

ERROR: glib-2.48 gthread-2.0 is required to compile QEMU

haodong@haodong-vm:~/Work/qemu-5.1.0$ sudo apt get install libglib2.0-dev
E: Invalid operation get
haodong@haodong-vm:~/Work/qemu-5.1.0$ sudo apt-get install libglib2.0-dev
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libblkid-dev libffi-dev libglib2.0-dev-bin libmount-dev libpcre16-3
  libpcre2-16-0 libpcre2-dev libpcre2-posix3 libpcre3-dev libpcre32-3
  libpcrecpp0v5 libselinux1-dev libsepol-dev python3-distutils uuid-dev zlib1g-dev
```

```

haodong@haodong-vm:~/Work/qemu-5.1.0$ ./configure --disable-kvm --disable-werror --p
refix=/usr/local --target-list="riscv64-softmmu"

ERROR: pixman >= 0.21.8 not present.
Please install the pixman devel package.

haodong@haodong-vm:~/Work/qemu-5.1.0$ sudo apt-get install libpixman-1-dev
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:

```

```

snappy support      no
bzip2 support       no
lzfs support        no
zstd support        no
NUMA host support   no
libxml2             no
tcmalloc support    no
jemalloc support    no
avx2 optimization   yes
avx512f optimization no
replication support yes
bochs support       yes
cloop support       yes
dmg support         yes
qcow v1 support     yes
vdi support         yes
vfat support        yes
qed support         yes
parallels support   yes
sheepdog support    yes
capstone            internal
libpmem support     no
libdaxctl support   no
libudev             no
default devices     yes
plugin support      no
fuzzing support     no
gdb                 /usr/bin/gdb-multiarch
rng-none            no
Linux keyring       yes
cross containers    no
haodong@haodong-vm:~/Work/qemu-5.1.0$

```

按照我的经验，失败三次后，第四次就会成功

(5) 输入 make 编译 QEMU，这一步较慢，中途可能出现一些 warning，不用理会

```

cross containers no
haodong@haodong-vm:~/Work/qemu-5.1.0$ make
GEN      riscv64-softmmu/config-devices.mak.tmp
GEN      riscv64-softmmu/config-devices.mak
GEN      config-all-devices.mak
GEN      config-host.h
make[1]: Entering directory '/home/haodong/Work/qemu-5.1.0/slrp'
GEN      /home/haodong/Work/qemu-5.1.0/slrp/src/libslrp-version

```

(6) 输入 sudo make install，至此环境配置完成


```

set -e; for x in da en-gb et fr fr-ch is lt no pt-br sv ar de
en-us fi fr-be hr it lv nl pl ru th de-ch es fo fr-ca
hu ja mk pt sl tr bepo cz; do \
install -c -m 0644 /home/haodong/Work/qemu-5.1.0/pc-bios/keymaps/$x "/usr/local/share/qemu/keymaps"; \
done
install -c -m 0644 /home/haodong/Work/qemu-5.1.0/trace-events-all "/usr/local/share/qemu/trace-events-all"
haodong@haodong-vm:~/Work/qemu-5.1.0$

```

3. 下载 xv6 源码

- (1) cd .. 退回 Work 目录, 此时 work 目录下只有 qemu-5.1.0
- (2) git clone git://g.csail.mit.edu/xv6-labs-2020 下载 xv6 代码

```

qemu/trace-events-all"
haodong@haodong-vm:~/Work/qemu-5.1.0$ cd ..
haodong@haodong-vm:~/Work$ git clone git://g.csail.mit.edu/xv6-labs-2020
Cloning into 'xv6-labs-2020'...
remote: Enumerating objects: 7008, done.
remote: Counting objects: 100% (7008/7008), done.
remote: Compressing objects: 100% (3612/3612), done.
remote: Total 7008 (delta 3650), reused 6526 (delta 3366), pack-reused 0
Receiving objects: 100% (7008/7008), 17.19 MiB | 5.17 MiB/s, done.
Resolving deltas: 100% (3650/3650), done.
haodong@haodong-vm:~/Work$

```

- (3) ls 查看 Work 目录, 应该有 qemu-5.1.0 和 xv6-labs-2020

```

haodong@haodong-vm:~/Work/qemu-5.1.0$ cd ..
haodong@haodong-vm:~/Work$ git clone git://g.csail.mit.edu/xv6-labs-2020
Cloning into 'xv6-labs-2020'...
remote: Enumerating objects: 7008, done.
remote: Counting objects: 100% (7008/7008), done.
remote: Compressing objects: 100% (3612/3612), done.
remote: Total 7008 (delta 3650), reused 6526 (delta 3366), pack-reused 0
Receiving objects: 100% (7008/7008), 17.19 MiB | 5.17 MiB/s, done.
Resolving deltas: 100% (3650/3650), done.
haodong@haodong-vm:~/Work$ ls
qemu-5.1.0  xv6-labs-2020
haodong@haodong-vm:~/Work$

```

五、测试环境是否正确

- (1) cd xv6-labs-2020 进入目录
- (2) git checkout util 切换到 util 分支
- (3) make qemu 通过 QEMU 运行 xv6, 如果出现下面的画面说明环境配置成功

```

riscv64-linux-gnu-ld -z max-page-size=4096 -N -e main -Text 0 -o user/_zombie user/
zombie.o user/_lib.o user/_sys.o user/_printf.o user/_malloc.o
riscv64-linux-gnu-objdump -S user/_zombie > user/_zombie.asm
riscv64-linux-gnu-objdump -t user/_zombie | sed '1,/SYMBOL TABLE/d; s/ .* //; /%5/d'
"> user/_zombie.syn
mkfs/mkfs fs.img README user/_argtest.sh user/_cat user/_echo user/_forktest user/
grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_str
essfs user/_usertests user/_grind user/_wc user/_zombie
mmeta 46 (boot, super, log blocks 30 inode blocks 13, bitmap blocks 1) blocks 954 to
tal 1000
ballocc: first 578 blocks have been allocated
ballocc: write bitmap block at sector 45
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -snp 3 -n
ographic -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive
=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$

```

(4) 先按 Ctrl + A , 之后按一下 X 就能退出 QEMU

如果最后运行时卡住, 大概率是因为某些软件版本不匹配

如果遇到 Ubuntu 网络问题等没提到的地方, 请自行上网查询或询问大模型来解决

环境配置完成后, 在 work 目录下创建一个你未来长期使用的目录(为你的内核取个名字吧, 比如 TOYOS, 注意设置为英文, 不要有特殊字符), 请将这个目录通过 git 同步到线上的 gitee 仓库。